VMamba-Mesh Study Report

Isomera Study Lab

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1 Purpose

This report records a VMamba-Mesh study/training run inside Isomera. VMamba is the reference backbone, and the Isomera adapter exports a benchmark-compatible pickle through a stable `predict_pairs(graph)` contract.

2 Source and Artifact Contract

Field	Value
Benchmark	smoke_operational
Scenario	graph_SOR2_D1_seed42
Graph	main/data/architectures/smoke_operational/gml/graph_SOR2_D1_seed42.gml
Labels	main/data/architectures/smoke_operational/real_pairs/graph_SOR2_D1_seed4
Pickle	main/data/architectures/smoke_operational/models/vmamba_mesh/VMambaM
Inference module	core.algorithms.vmamba_mesh

3 VMamba to VMamba-Mesh Delta

Component	VMamba	VMamba-Mesh
Input	Natural image patches	Canonical lineage pair tensors
Scan	SS2D over image rows/columns	Lineage-aware scan over SOR–SOT–SPEC blocks
Identity	Pixel channels	DiagFP schema/table fingerprint channel
State	Visual context memory	HierInit upstream lineage memory
Sparsity	Dense image assumption	SparseGate for sparse adjacency matrices
Output	Image class/logits	Duplicate-pair score and benchmarkable pair set

4 Training Summary

Metric	Value
Model version	vmamba_mesh_isomera_adapter_v0.1
Dataset pairs	5
Positive pairs	1
Negative pairs	4
Selected threshold	0.625
Calibration Jaccard	1.0
Calibration accuracy	1.0
Calibration precision	1.0
Calibration recall	1.0

5 Reproducibility

The saved pickle expects a directed `networkx.DiGraph` with lineage table nodes and returns predicted duplicate pairs as a list of string tuples. The benchmark layer computes TP, FP, FN, TN, Jaccard, SF-Jaccard and ET using the same evaluator used by VF2, Node Match and GNN pickle clusters.